does not include speaker information. Instead, the sound is represented in a speaker independent format called the B-format, where the sound is decoded based on the number of speakers before playback. This provides a lot of flexibility in terms of VR where sound can possibly originate from anywhere.

In 3D development, it is possible to place the sound source within an object where the sound is supposed to originate. Going one step further, current technology allows us to create motion in such sound sources, which is somewhat tedious as it has to be done manually, but achieves the effect of the sound moving with the object it is supposed to belong to. Head tracking within headsets like Oculus can actually take care of collating the movement of the user with respect to the virtual audio source. Another interesting approach is to programmatically trigger sounds based on where the user is looking. Although it does not link to sound's behavior in reality, it might provide some interesting use cases, like triggering a specific narration when a particular object is observed in a virtual museum. One of the most significant companies that is working in this space right now is Thrive Audio, that is working on positional sound and was acquired by Google in early 2015.

Adding dimensions

Along with the need for new and fresh VR content, the advent of VR has created a desire amid people to see their favorite movies or experience their favorite games all over in VR. Upscaling 2D content to 3D has been around for a while, but only for certain specific cases. The task basically involves taking a 2D image and creating one image for each eye out of it. But the real work is not as easy as it sounds. The quality of the 2D image has to be maintained and even improved in some cases to maintain immersion. The major difference in experience is that native 3D seems to pop out of the screen whereas 2D-3D goes deeper into the screen.

Upscaling 2D to 3D requires foreground and background detection along with object identification. These area are implemented using algorithms. There are multiple approaches to this:

- Re-use the 3D design files to generate the entire experience in 3D
- Depth map based approach

The first approach can only be taken if there is a pre-existing 3D based development files, which might not be true all the time, and definitely not that useful.

In the second technique, a Separate grayscale image for each frame is generated where each depth is shaded according to its detected depth